

2.Aim

To verify the **integrity of digital evidence** by generating and comparing the SHA-256 hash value of a file.

Algorithm

1. Start the program.
2. Select the file to be verified.
3. Read the file in binary mode.
4. Generate the SHA-256 hash of the file.
5. Generate the hash again for verification.
6. Compare the original hash with the verification hash.
7. If both hashes are the same, display **"Chain of Custody Verified."**
8. Otherwise, display **"File Modified."**
9. Stop the program.

Code

```
import hashlib
```

```
import os
```

```
def calculate_hash(file_path):
```

```
    hash_sha256 = hashlib.sha256()
```

```
    with open(file_path, "rb") as f:
```

```
        while True:
```

```
            chunk = f.read(4096)
```

```
            if not chunk:
```

```
                break
```

```
            hash_sha256.update(chunk)
```

```
    return hash_sha256.hexdigest()
```

```
file = "sample.txt"

# Create the file if it doesn't exist
if not os.path.exists(file):
    with open(file, "w") as f:
        f.write("This is digital evidence.")

original_hash = calculate_hash(file)
print("Original Hash:", original_hash)

verification_hash = calculate_hash(file)
print("Verification Hash:", verification_hash)

if original_hash == verification_hash:
    print("✅ Chain of Custody Verified")
else:
    print("❌ File Modified")
```

Output

Original Hash:

a1c7197000c38e802f2c71b3a2d2700c38335cb74863dd192cbd7eb93f18fe08

Verification Hash:

a1c7197000c38e802f2c71b3a2d2700c38335cb74863dd192cbd7eb93f18fe08

✅ Chain of Custody Verified